

Covariant Local Jets of Response-Matched Neutral-Atom Controls: From Quartic Prediction to Formal Finite-Error Certification

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Abstract

Control sequences can agree at their ideal endpoint and in their complete first-order response to specified implementation errors while differing at finite error. We study this phenomenon in a two-atom Rydberg model with piecewise-constant phase control. Twenty-four control phases are constrained to match the output state and its first-order state response to global amplitude, detuning, and interaction errors. The resulting numerical constraint Jacobian has rank sixteen, leaving an eight-dimensional implementation fibre. Paths on this fibre display distinct finite-error infidelities.

We first identify a fourth-order symmetric response coefficient, $\mathcal{G}_4$, as a strong low-order predictor of mean finite-error performance. In a protocol-frozen prospective twenty-path validation it attains Spearman correlation 0.998496 and predicts the best path correctly. We then recover the full three-dimensional symmetric fourth-order tensor. Under a nonorthogonal change of error coordinates, direct recomputation agrees with the tensor transformation law to 1.1×10^{-14} , while contraction with the transformed fourth noise moment is invariant to 2.0×10^{-15} .

The quartic term is not universally sufficient. We therefore compute the zero-error Taylor jet through order thirty by block-matrix exponentials and bound the uncomputed tail analytically. Finally, a 192-bit Arb midpoint-radius calculation uses outward-rounded complex balls, a 64-point Cauchy extraction, a rigorous alias enclosure, and an analytic order-32 tail enclosure. For a frozen twelve-path cohort, the quartic term alone certifies 35 of 66 path pairs, whereas the order-thirty jet certifies all 66 pairs. All ordinary held-out outcomes lie inside the formal intervals.

The result is a formal ordering certificate for a serialized-decimal, finite-dimensional Hamiltonian model. It is not evidence from PASQAL hardware, and exact existence of the numerically matched implementation fibre remains to be certified by an interval-Newton or Krawczyk argument.

1 Introduction

Pulse-level control of programmable neutral-atom arrays provides direct access to Rabi amplitude, phase, detuning, geometry, and Rydberg-mediated interactions [1, 2, 3]. This flexibility also creates a basic identification problem. Two implementations may realize the same ideal state and have the same leading response to calibrated errors, yet behave differently when the errors are finite. Endpoint agreement alone does not identify an implementation, and even first-order robustness data may leave a nontrivial family of controls.

Composite-pulse and robust-control methods commonly cancel or suppress leading error terms [4, 5, 6]. The question asked here is different:

After the ideal output state and the complete first-order output-state response have been matched, what local object predicts the residual finite-error ordering inside the remaining implementation fibre?

We address the question in four stages.

1. We construct phase schedules with a common endpoint and common first-order state response to amplitude, detuning, and Rydberg-interaction errors.
2. We test a low-order quartic response scalar, $\mathcal{G}_4$, using a frozen ranking followed by held-out finite-error evaluation.
3. We promote $\mathcal{G}_4$ to a coordinate-covariant fourth-order tensor whose contraction with a physical noise moment is invariant.
4. We compute higher local derivatives and produce an outward-rounded Arb ball-arithmetic certificate of finite-error ordering.

The central outcome is deliberately narrower than a universal robustness law. The fourth-order term is often predictive and can certify a substantial subset of pairwise comparisons, but close paths require higher derivatives. For the serialized model studied here, the order-thirty local jet plus an analytic tail bound certifies the complete finite-error ordering. This distinguishes three statements that are often conflated:

low-order correlation $\neq$ partial pairwise certification $\neq$ complete bounded reconstruction.

2 Two-atom control model

2.1 Hamiltonian

We use the ordered basis

$$\{|gg\rangle, |gr\rangle, |rg\rangle, |rr\rangle\}.$$

Let

$$\begin{aligned} X_\Sigma &= X \otimes I + I \otimes X, & Y_\Sigma &= Y \otimes I + I \otimes Y, \\ N_\Sigma &= n \otimes I + I \otimes n, & N_{rr} &= n \otimes n, \end{aligned}$$

where $n = |r\rangle\langle r|$. During segment j , the Hamiltonian is

$$\begin{aligned} H_j(\epsilon) &= \frac{\Omega}{2}(1 + \epsilon_\Omega)(\cos \phi_j X_\Sigma + \sin \phi_j Y_\Sigma) \\ &\quad - \Omega \epsilon_\Delta N_\Sigma + V(1 + \epsilon_V)N_{rr}. \end{aligned} \tag{1}$$

The dimensionless error vector is

$$\epsilon = (\epsilon_\Omega, \epsilon_\Delta, \epsilon_V).$$

The nominal parameters are

$$\Omega = 2\pi \text{ rad}/\mu\text{s}, \quad V = 4\Omega,$$

with 24 segments of duration

$$\tau = 0.1 \mu\text{s}, \quad T = 24\tau = 2.4 \mu\text{s}.$$

The nominal interaction can be represented as $V = C_6/R^6$; using the Pulser DigitalAnalogDevice coefficient in the accompanying implementation gives a nominal separation of approximately $7.744 \mu\text{m}$.

For a control path

$$\gamma = (\phi_1, \dots, \phi_{24}) \in \mathbb{T}^{24},$$

the output state is

$$|\psi_\gamma(\epsilon)\rangle = \prod_{j=24}^1 \exp[-i\tau H_j(\epsilon)] |gg\rangle. \quad (2)$$

All primary calculations use exact matrix exponentials for the piecewise constant, finite-dimensional model. Pulser/QuTipEmulator was used in an earlier two-atom cross-check, but the formal certificate in this work is independent of Pulser and does not invoke a cloud backend.

2.2 Loss and held-out error law

Let γ_{ref} denote the reference schedule and define

$$|\psi_{\text{ref}}\rangle = |\psi_{\gamma_{\text{ref}}}(\mathbf{0})\rangle.$$

The state infidelity is

$$\mathcal{I}_\gamma(\epsilon) = 1 - |\langle\psi_{\text{ref}}|\psi_\gamma(\epsilon)\rangle|^2. \quad (3)$$

The held-out error design contains the six signed axial errors

$$\mathcal{E} = \{\pm 0.06 e_\Omega, \pm 0.04 e_\Delta, \pm 0.05 e_V\}. \quad (4)$$

The primary performance functional is the mean

$$\bar{\mathcal{I}}_\gamma = \frac{1}{6} \sum_{\epsilon \in \mathcal{E}} \mathcal{I}_\gamma(\epsilon). \quad (5)$$

Worst-case infidelity is retained as a secondary diagnostic. The present predictors do not reliably rank the worst case, and no worst-case theorem is claimed.

3 Matched implementation fibre

3.1 Endpoint and first-order constraints

At zero error, remove the unobservable global phase by the horizontal projector

$$P_\perp = I - |\psi_0\rangle\langle\psi_0|.$$

Define the horizontal first-order responses

$$J_{\gamma,a} = P_\perp \left. \frac{\partial}{\partial \epsilon_a} |\psi_\gamma(\epsilon)\rangle \right|_{\epsilon=0}, \quad a \in \{\Omega, \Delta, V\}. \quad (6)$$

Definition 1 (Numerically matched implementation fibre). *For a declared tolerance η , the numerical matched fibre $\mathcal{F}_\eta$ consists of controls γ satisfying*

$$\begin{aligned} \mathcal{I}_\gamma(\mathbf{0}) &\leq \eta, \\ \|J_{\gamma,a} - J_{\text{ref},a}\| &\leq \eta, \quad a \in \{\Omega, \Delta, V\}. \end{aligned} \quad (7)$$

The actual optimizer retains the full phase-aligned complex output state and the three full phase-aligned complex state tangents. Redundant real components are retained in the residual vector, while singular-value decomposition determines the independent rank.

3.2 Dimension count and generation

The control has 24 phase variables. At the reference solution, the numerical constraint Jacobian has rank

$$\text{rank } DC(\gamma_{\text{ref}}) = 16,$$

leaving an eight-dimensional numerical null space. Candidate paths are generated by:

1. drawing a seeded direction in the numerical null space;
2. applying a finite displacement;
3. projecting back to the constraint set by nonlinear least squares;
4. rejecting candidates whose constraint residual or pairwise phase distance fails a predeclared gate.

No held-out finite-error outcome is evaluated during candidate generation. The maximum constraint residual in the reported runs is of order 10^{-11} .

Remark 1 (Unfinished exact-fibre step). *The numerical rank and residual establish a highly accurate computational fibre, not the exact existence of an eight-dimensional solution manifold. A complete exact-fibre theorem requires a square transverse formulation and an interval-Newton or Krawczyk inclusion proof. This limitation is kept separate from the formal ordering certificate below.*

4 Local response hierarchy

4.1 Symmetric local response

Introduce normalized error coordinates $z \in \mathbb{R}^3$ through

$$\epsilon = Dz, \quad D = \text{diag}(0.06, 0.04, 0.05).$$

For a direction $v \in \mathbb{R}^3$, define the symmetric response

$$S_\gamma(t, v) = \frac{\mathcal{I}_\gamma(tDv) + \mathcal{I}_\gamma(-tDv)}{2}. \quad (8)$$

Analyticity of the finite matrix exponential gives

$$S_\gamma(t, v) = q_{2,\gamma}(v)t^2 + q_{4,\gamma}(v)t^4 + q_{6,\gamma}(v)t^6 + \dots. \quad (9)$$

Odd powers cancel by construction.

First-order output-state matching fixes the leading quadratic loss to numerical precision. Across the validation cohorts, the relative spread of the averaged quadratic coefficient is approximately 10^{-7} . The first useful discriminator is therefore the quartic coefficient.

4.2 Quartic tensor

There exists a symmetric fourth-order tensor $A_\gamma^{(4)}$ such that

$$q_{4,\gamma}(v) = A_{\gamma,ijkl}^{(4)} v_i v_j v_k v_l. \quad (10)$$

In three dimensions a symmetric fourth-order tensor has

$$\binom{3+4-1}{4} = 15$$

independent components. We recover them from 36 seeded directions. The directional Taylor coefficients are computed directly at $t = 0$ through order six by block-matrix exponentials, rather than by fitting nonzero-radius samples.

For the six-axis law in Eq. (4), define the fourth moment

$$M_{ijkl}^{(4)} = \mathbb{E}[z_i z_j z_k z_l].$$

The scalar quartic predictor is

$$\mathcal{G}_4(\gamma; M^{(4)}) = A_{\gamma,ijkl}^{(4)} M_{ijkl}^{(4)}. \quad (11)$$

For the equally weighted signed axes,

$$\mathcal{G}_4 = \frac{1}{3} \left(A_{\Omega\Omega\Omega\Omega}^{(4)} + A_{\Delta\Delta\Delta\Delta}^{(4)} + A_{VVVV}^{(4)} \right).$$

4.3 Coordinate covariance

Proposition 1 (Covariance and invariant contraction). *Let $z = Ry$ for an invertible real matrix R . Then*

$$A_{y,abcd}^{(4)} = A_{z,ijkl}^{(4)} R_{ia} R_{jb} R_{kc} R_{ld}, \quad (12)$$

$$M_{y,abcd}^{(4)} = (R^{-1})_{ai} (R^{-1})_{bj} (R^{-1})_{ck} (R^{-1})_{dl} M_{z,ijkl}^{(4)}, \quad (13)$$

and

$$A_y^{(4)} : M_y^{(4)} = A_z^{(4)} : M_z^{(4)}. \quad (14)$$

Proof. Equation (12) follows by substituting $z = Ry$ into the quartic form. Equation (13) follows from $y = R^{-1}z$. Contracting all four indices cancels each R with the corresponding R^{-1} , proving Eq. (14). $\square$

This proposition is algebraic. The numerical audit additionally recomputes the tensor directly in the transformed coordinates. For a nonorthogonal map of condition number approximately 1.82, the direct-refit relative error is 1.08×10^{-14} and the maximum invariant-contraction relative error is 1.99×10^{-15} .

5 Prospective quartic ranking

5.1 Discovery and validation separation

The initial four named controls were used only for discovery and orientation. A twelve-path cohort was then used to compare candidate formulas. The smallest sufficient rule on that training cohort was

$$\text{smaller } \mathcal{G}_4 \implies \text{smaller mean finite-error infidelity.}$$

The addition of the sixth-order coefficient did not improve the training rank correlation, so the quartic rule was frozen.

A new twenty-path cohort was generated with a different seed. For each candidate, a ranking certificate containing phases, local coefficients, and predicted order was written and SHA-256 hashed before the held-out outcome evaluator was enabled. The predeclared primary gate was

$$\rho_{\text{Spearman}} \geq 0.95,$$

together with correct top-path identification.

5.2 Quartic validation result

The twenty-path Colab run gave

$$\rho_{\text{Spearman}}(-\mathcal{G}_4, -\bar{\mathcal{I}}) = 0.998496, \quad p = 3.71 \times 10^{-24}. \quad (15)$$

The predicted and observed top path agreed. Only one adjacent pair was inverted in the complete ranking. Two local coefficient-extraction windows had rank correlation one, and the maximum local fit residual was 1.15×10^{-10} .

Subsequent cohorts show that $\mathcal{G}_4$ is not universally sufficient. Depending on the generated matched paths, prospective quartic Spearman correlations ranged from approximately 0.85 to 1.00. This variation motivated the bounded higher-order analysis rather than post hoc replacement of the quartic formula.

6 High-order local reconstruction

6.1 Exact zero-point coefficients

For one segment, write

$$U_j(t) = \exp(A_j + tB_j), \quad A_j = -i\tau H_j(0).$$

The coefficient of t^n can be obtained from the $(0, n)$ block of the exponential of an upper-bidiagonal block matrix with A_j on the diagonal and B_j on the first superdiagonal. Segment series are multiplied and truncated to order $N = 30$. This produces the zero-point state jet without finite-radius differencing.

For the symmetric mean loss,

$$\bar{\mathcal{I}}_\gamma(t) = C_{0,\gamma} + C_{2,\gamma}t^2 + \mathcal{G}_{4,\gamma}t^4 + \sum_{m=3}^{15} \mathcal{G}_{2m,\gamma}t^{2m} + \mathcal{R}_{32,\gamma}(t). \quad (16)$$

The small C_0 term is retained because the numerical candidate endpoint is not symbolically identical to the reference endpoint.

6.2 Analytic tail bound

Let K_a be the integrated spectral norm of the Hamiltonian perturbation along axis a :

$$K_a = \int_0^T \|\partial_t H_a(t)\|_2 dt.$$

For the normalized held-out axes,

$$(K_\Omega, K_\Delta, K_V) \approx (0.90478, 1.20637, 3.01593).$$

A Dyson-series norm estimate yields

$$|\mathcal{R}_{32,\gamma}(1)| \leq \frac{1}{3} \sum_{a \in \{\Omega, \Delta, V\}} \sum_{\substack{n \geq 32 \\ n \text{ even}}} \frac{(2K_a)^n}{n!}. \quad (17)$$

The floating-point order-thirty audit evaluates the right-hand side as approximately 1.2343×10^{-11} . The maximum difference between the order-thirty reconstruction and ordinary held-out propagation was below 10^{-15} in the reported runs.

6.3 Two notions of certification

To test the quartic term alone, define the quartic centre

$$\mu_\gamma^{(4)} = \overline{C}_0 + \overline{C}_2 + \mathcal{G}_{4,\gamma},$$

and place the complete known correction

$$(C_{0,\gamma} - \overline{C}_0) + (C_{2,\gamma} - \overline{C}_2) + \sum_{m=3}^{15} \mathcal{G}_{2m,\gamma}$$

inside the interval radius together with the tail bound. Two paths are quartic-certified only if these conservative intervals are disjoint.

For the order-thirty certificate, the centre includes all known terms through order thirty and the radius contains only the analytic tail and verified rounding enclosure.

7 Formal Arb ball certificate

7.1 Outward-rounded coefficient extraction

Arb implements arbitrary-precision midpoint-radius arithmetic with automatic error tracking [7]. The final audit uses 192-bit real and complex balls. Instead of exponentiating large block matrices in ball arithmetic, amplitude coefficients are extracted by a 64-point Cauchy discrete Fourier transform on the complex unit circle:

$$\hat{g}_n = \frac{1}{64} \sum_{k=0}^{63} g(\zeta_k) \zeta_k^{-n}, \quad \zeta_k = e^{2\pi i k/64}. \quad (18)$$

The discrete transform aliases coefficients $g_{n+64\ell}$. A path-uniform Dyson bound encloses this alias. The implemented enclosure is 10^{-40} , and the formal gate verifies that

$$e^{K_a} \frac{K_a^{64}}{64!} < 10^{-40}$$

for every physical error axis. Fidelity coefficients are then formed by ball convolution,

$$F_n = \sum_{r=0}^n \overline{g}_r g_{n-r},$$

and the infidelity coefficients follow from $1 - F$.

7.2 Conditional formal theorem

Theorem 1 (Certified ordering for the serialized model). *Consider the twelve controls serialized in the formal certificate with protocol hash*

`2c27f222ff6bcb7e87f9073da1b3f151`
`9c08498124e772796d363cab8f5c5338.`

For the finite-dimensional Hamiltonian in Eq. (1), the common reference target, and the six-axis error law in Eq. (4), the 192-bit Arb order-thirty intervals are pairwise disjoint and certify the complete ordering of all 66 unordered path pairs.

Table 1: Representative prospective and formal audits. “Pairs” denotes certified unordered comparisons. Candidate sets differ between stages and between numerical environments; every stage freezes its protocol and ranking before held-out evaluation.

Audit	Paths	Spearman ρ	Certified pairs	Result
Quartic prospective validation	20	0.998496	–	pass
Covariant tensor validation	12	1.000000	–	pass
Order-30 floating certificate	12	1.000000	66/66	pass
Arb formal: quartic only	12	–	35/66	partial
Arb formal: order 30	12	1.000000	66/66	pass

Table 2: Numerical integrity diagnostics from representative Colab runs.

Diagnostic	Value
Fourth-tensor independent components	15
Maximum tensor recovery relative residual	4.16×10^{-15}
Direct coordinate-refit relative error	1.08×10^{-14}
Invariant-contraction relative error	1.99×10^{-15}
Quadratic-coefficient relative spread	1.33×10^{-7}
Arb precision	192 bits
Cauchy points	64
Taylor order	30
Formal quartic pair coverage	$35/66 = 53.03\%$
Formal order-30 pair coverage	$66/66 = 100\%$

Computer-assisted proof. All pulse phases and constants are parsed as serialized decimals or exact rational multiples of the Arb enclosure of π . Arb complex balls enclose every trigonometric evaluation, matrix exponential, propagation, Cauchy sum, and coefficient convolution. The Cauchy alias gate proves that the omitted Fourier aliases lie inside the declared 10^{-40} complex enclosure. The analytic even Dyson tail after order thirty is enclosed by 2×10^{-11} . The resulting 12 order-thirty real balls are pairwise disjoint in the order recorded in the pre-outcome certificate. Direct interval comparisons certify all $\binom{12}{2} = 66$ pairs. The certificate is written and hashed before ordinary held-out outcomes are evaluated. Finally, every ordinary outcome is verified to lie inside its corresponding formal ball. $\square$

Remark 2 (Scope of Theorem 1). *The theorem is conditional on the serialized-decimal finite Hamiltonian model and the listed candidate controls. It does not certify model discrepancy, calibration drift, sampling noise, or PASQAL hardware. It also does not prove that the numerical paths lie on an exact first-order matched fibre; it proves the finite-error ordering of those serialized controls.*

8 Results

The sequence of results supports the following hierarchy.

1. **Implementation memory:** endpoint and complete first-order output-state matching do not numerically determine finite-error performance.
2. **Low-order organization:** the covariant quartic contraction $\mathcal{G}_4$ strongly organizes mean performance and formally certifies a substantial subset of path pairs.

3. **High-order closure:** close comparisons depend on higher local derivatives. The order-thirty local jet plus a bounded tail certifies the complete ordering for the serialized cohort.

9 Discussion

9.1 What is geometrical?

The geometrical object is not the scalar $\mathcal{G}_4$ in isolation. It is the fourth-order response tensor $A^{(4)}$ together with the physical fourth noise moment $M^{(4)}$. Their contraction is independent of a linear reparameterization of error coordinates. This avoids interpreting a coordinate-specific axis average as an intrinsic curvature.

The tensor should also not be identified automatically with Riemann curvature. It is a quartic susceptibility of the endpoint loss—a component of the fourth jet of the endpoint map composed with infidelity. Establishing a relation to a canonical connection or curvature on a control quotient would require additional structure.

9.2 Why the quartic term is useful but incomplete

First-order matching makes the quadratic loss nearly common. This promotes the quartic term to the first visible discriminator. It explains why $\mathcal{G}_4$ often gives almost perfect rankings. However, if two quartic coefficients are close, the sixth and higher terms can reverse their order. The formal intervals expose this distinction directly: 35 pairs are quartic-certified, while the remaining pairs require higher jet data.

This is preferable to declaring the quartic rule either universally true or false. The correct statement is gap-dependent:

$$|\mathcal{G}_{4,a} - \mathcal{G}_{4,b}| > B_a^{(\geq 6)} + B_b^{(\geq 6)} \implies \text{quartic ordering is certified.} \quad (19)$$

9.3 Relation to analyticity

An analytic finite-dimensional propagator is determined by its full local germ, so the existence of some sufficiently high reconstructing order is not by itself surprising. The nontrivial computational content is quantitative:

- a finite order, 30, suffices at the full held-out scale;
- the omitted tail is explicitly bounded;
- all pairwise intervals are disjoint;
- a much lower fourth-order object already certifies a substantial fraction of comparisons.

Thus the result is not that “analytic functions have Taylor series,” but that a finite, auditable local resource closes a nontrivial control-ranking problem.

9.4 Implications for control selection

For a noise moment M , the low-cost screening problem is

$$\gamma^* \in \arg \min_{\gamma \in \mathcal{F}_\eta} A_\gamma^{(4)} : M^{(4)}.$$

Pairs that fail the quartic gap test in Eq. (19) can be escalated adaptively to sixth, eighth, or higher jet order until their intervals separate. This suggests a certified branch-and-bound control selector: compute low-order local tensors for all candidates, refine only ambiguous pairs, and reserve direct finite-error simulation or hardware evaluation for model validation rather than exhaustive ranking.

10 Limitations and next theorem

The present work has five explicit limitations.

1. **Exact-fibre existence.** The matching constraints are satisfied to numerical tolerance. A formal interval-Newton/Krawczyk proof has not yet established a nearby exact matched solution for every candidate.
2. **Finite-dimensional model.** Spontaneous emission, Doppler effects, laser phase noise, position fluctuations, leakage levels, and waveform filtering are absent.
3. **Two atoms.** No many-body scaling theorem is claimed.
4. **Mean axial error law.** The primary result concerns the declared six-axis mean. Worst-case ranking is not supported by the present scalar.
5. **No QPU evidence.** The calculations are local and require no PASQAL credentials. They do not constitute PASQAL Cloud, FRESNEL, or physical-QPU evidence.

The next mathematical step is to decompose the 24-dimensional phase space into sixteen transverse variables and eight fibre variables, formulate an independent analytic constraint map, and prove a Krawczyk inclusion

$$K(X) \subset \text{int}(X)$$

for a box X around each numerical path. Propagating the resulting phase balls through the formal jet calculation would close the chain

$$\text{exact matched control} + \text{bounded local jet} \implies \text{finite-error ordering.}$$

11 Reproducibility

The principal standalone scripts are:

- `pasqal_two_atom_G4_standalone_colab.py`;
- `pasqal_L3_L4_standalone_colab.py`;
- `pasqal_L4_order30_standalone_colab.py`;
- `pasqal_L4_formal_arb_standalone_colab.py`.

The final formal protocol hash is

```
2c27f222ff6bcb7e87f9073da1b3f151
9c08498124e772796d363cab8f5c5338.
```

The certificate hash is run-specific because the certificate contains a creation time and the actual accepted path realization. The formal standalone script pins `python-flint==0.8.0`; NumPy and SciPy are used for path generation, while Arb/ACB is used for the formal interval certificate.

12 Conclusion

We have separated finite-error robustness inside a numerically matched neutral-atom implementation fibre into a hierarchy of local information. The common endpoint and complete first-order state response leave residual path dependence. A coordinate-covariant fourth-order response tensor provides a strong and partially certifying low-order description. Close comparisons require higher derivatives. For a frozen twelve-path serialized model, the zero-error jet through

order thirty, combined with rigorous Cauchy-alias and Dyson-tail enclosures, yields an Arb outward-rounded certificate of all 66 finite-error pairwise orderings.

The strongest justified conclusion is therefore conditional but precise: finite-error ordering in this model is recoverable from a finite local jet with a formal remainder enclosure. Establishing exact matched-fibre existence and extending the certificate to open-system, many-atom, and hardware data are the remaining steps.

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